

Matrices & Vectors

ANSWER KEY



Addition and scalar multiplication

- 1 Let A be the matrix with rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$, and B be the matrix with rows $\langle 2, -1 \rangle$ and $\langle 0, 5 \rangle$. Find $A + B$ and $2A$.

$A + B$ has rows $\langle 3, 2 \rangle$ and $\langle -2, 9 \rangle$. $2A$ has rows $\langle 2, 6 \rangle$ and $\langle -4, 8 \rangle$.

Matrix multiplication

- 2 Using A (rows $\langle 1, 3 \rangle$, $\langle -2, 4 \rangle$) and B (rows $\langle 2, -1 \rangle$, $\langle 0, 5 \rangle$) from above, find AB .

AB has rows $\langle 2, 14 \rangle$ and $\langle -4, 22 \rangle$: e.g. row 1, column 1 is $(1)(2) + (3)(0) = 2$.

- 3 Let A have rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$, and let $\mathbf{v} = \langle 3, -1 \rangle$. Find $A\mathbf{v}$.

$A\mathbf{v} = \langle (1)(3) + (3)(-1), (-2)(3) + (4)(-1) \rangle = \langle 0, -10 \rangle$.

Determinant and inverse

- 4 Find $\det A$ for the matrix A with rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$, and state whether A is invertible.

$\det A = (1)(4) - (3)(-2) = 4 + 6 = 10 \neq 0$, so A is invertible.

- 5 Find A^{-1} for the matrix A with rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$.

$A^{-1} = \frac{1}{10} \times$ the matrix with rows $\langle 4, -3 \rangle$ and $\langle 2, 1 \rangle$, i.e., A^{-1} has rows $\langle 0.4, -0.3 \rangle$ and $\langle 0.2, 0.1 \rangle$.

Solving systems with the inverse matrix

- 6 Solve the system $x + 3y = 13$ and $-2x + 4y = 4$ using $\mathbf{x} = A^{-1}\mathbf{b}$ (using A^{-1} from above).

A^{-1} has rows $\langle 0.4, -0.3 \rangle$ and $\langle 0.2, 0.1 \rangle$. Then $x = 0.4(13) - 0.3(4) = 5.2 - 1.2 = 4$ and $y = 0.2(13) + 0.1(4) = 2.6 + 0.4 = 3$. Check: $4 + 3(3) = 13$ and $-2(4) + 4(3) = 4$.

Vector magnitude and dot product

- 7 For $\mathbf{u} = \langle 4, 3 \rangle$ and $\mathbf{w} = \langle -3, 4 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

$\mathbf{u} \cdot \mathbf{w} = (4)(-3) + (3)(4) = 0$, and $|\mathbf{u}| = \sqrt{4^2 + 3^2} = 5$.

Angle between vectors

- 8 Find the angle between $\mathbf{p} = \langle 2, 2 \rangle$ and $\mathbf{q} = \langle 3, 0 \rangle$.

$$\mathbf{p} \cdot \mathbf{q} = 6, |\mathbf{p}| = 2\sqrt{2}, |\mathbf{q}| = 3. \cos \theta = \frac{6}{6\sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.7071 \Rightarrow \theta = 45^\circ.$$

- 9 Find the angle between $\langle 7, 1 \rangle$ and $\langle 2, -3 \rangle$, rounded to the nearest tenth of a degree.

$$\text{Dot product} = 14 - 3 = 11, \text{ magnitudes } \sqrt{50} \text{ and } \sqrt{13}. \cos \theta = \frac{11}{\sqrt{650}} \approx 0.4314 \Rightarrow \theta \approx 64.4^\circ.$$